

2014-2015 – Master 2 – Macro I

Complement for problem set 4 : Computing the
speed of convergence

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1. The setup

- Production is Cobb-Douglas :

$$Y = AK^{\alpha}. \quad (1)$$

- Capital accumulation is

$$\dot{K} = sY - \delta K$$

2. With constant output level in the long run

- ▶ Assume that A is constant.
- ▶ In the long run the economy converges to a steady state with output and capital given by Y_{LR} and K_{LR} , respectively.
- ▶ The output gap (distance from SS) is given by

$$\hat{y} = \frac{Y - Y_{LR}}{Y_{LR}}.$$

- ▶ So we have

$$Y = Y_{LR}(1 + \hat{y}).$$

- ▶ We can also see that

$$\ln \frac{Y}{Y_{LR}} = \ln(1 + \hat{y}) \approx \hat{y}.$$

- ▶ We can similarly define $\hat{k} = \frac{K - K_{LR}}{K_{LR}} = \ln \frac{K}{K_{LR}}$.
- ▶ Taking logs in (1) yields

$$\hat{y} = \alpha \hat{k}.$$

2. With constant output level in the long run

- By definition, the speed of convergence is given by

$$v = -\frac{1}{\hat{y}} \frac{d\hat{y}}{dt}.$$

- We have that

$$\begin{aligned} \frac{d\hat{y}}{dt} &= \frac{d}{dt} \ln \frac{Y}{Y_{LR}} \\ &= \frac{\dot{Y}}{Y} = \alpha \frac{\dot{K}}{K} \\ &= \alpha \left(\frac{sY - \delta K}{K} \right) \\ &= \alpha (sAK^{\alpha-1} - \delta) \\ &= \alpha (sAK_{LR}^{\alpha-1} (1 + \hat{k})^{\alpha-1} - \delta). \end{aligned}$$

2. With constant output level in the long run

- ▶ For $x \ll 1$ a first order approximation is $(1 + x)^n = 1 + nx$.
- ▶ Therefore,

$$\begin{aligned}\frac{d\hat{y}}{dt} &= \alpha(sAK_{LR}^{\alpha-1}(1 + \hat{k})^{\alpha-1} - \delta) \\ &= \alpha(sAK_{LR}^{\alpha-1}(1 + (\alpha - 1)\hat{k}) - \delta) \\ &= \alpha(sAK_{LR}^{\alpha-1} - \delta) + \alpha(\alpha - 1)sAK_{LR}^{\alpha-1}\hat{k}.\end{aligned}$$

2. With constant output level in the long run

- But in steady state,

$$\begin{aligned}\dot{K} &= 0 \\ \implies sY_{LR} - \delta K_{LR} &= 0 \\ \implies sAK_{LR}^{\alpha} &= \delta K_{LR} \\ \implies sAK_{LR}^{\alpha-1} &= \delta.\end{aligned}$$

- Consequently,

$$\begin{aligned}\frac{d\hat{y}}{dt} &= \alpha(sAK_{LR}^{\alpha-1} - \delta) + \alpha(\alpha - 1)sAK_{LR}^{\alpha-1}\hat{k} \\ &= \alpha(\alpha - 1)\delta\hat{k} \\ &= (\alpha - 1)\delta\hat{y}.\end{aligned}$$

- Thus,

$$v = \delta(1 - \alpha).$$

3. With long run growth

- ▶ Let us now assume that A grows at a constant rate g_A , so that both Y and K grow at the common constant rate g in the long run.
- ▶ Using the production function, we can check that we must have

$$\frac{\dot{Y}}{Y} = \frac{\dot{A}}{A} + \alpha \frac{\dot{K}}{K},$$

- ▶ i.e., in a balanced growth path,

$$\begin{aligned} g &= g_A + \alpha g, \\ \iff g &= \frac{g_A}{1 - \alpha}. \end{aligned}$$

3. With long run growth

- Our computations must be modified as follows :

$$\begin{aligned}\frac{d\hat{y}}{dt} &= \frac{d}{dt} \ln \frac{Y}{Y_{LR}} \\&= \frac{\dot{Y}}{Y} - \frac{\dot{Y}_{LR}}{Y_{LR}} = g_A + \alpha \frac{\dot{K}}{K} - g \\&= \alpha \left(\frac{\dot{K}}{K} - g \right) \\&= \alpha \left(\frac{sY - \delta K}{K} - g \right) \\&= \alpha (sAK^{\alpha-1} - \delta - g) \\&= \alpha (sAK_{LR}^{\alpha-1} (1 + \hat{k})^{\alpha-1} - \delta - g) \\&= \alpha (sAK_{LR}^{\alpha-1} (1 + (\alpha - 1)\hat{k}) - \delta - g) \\&= \alpha (sAK_{LR}^{\alpha-1} - \delta - g) + \alpha(\alpha - 1)sAK_{LR}^{\alpha-1}\hat{k}.\end{aligned}$$

3. With long run growth

- In steady state,

$$\begin{aligned}\dot{K} &= gK \\ \implies sY_{LR} - \delta K_{LR} &= gK_{LR} \\ \implies sAK_{LR}^{\alpha} &= (g + \delta)K_{LR} \\ \implies sAK_{LR}^{\alpha-1} &= g + \delta.\end{aligned}$$

- Thus,

$$\begin{aligned}\frac{d\hat{y}}{dt} &= \alpha(sAK_{LR}^{\alpha-1} - \delta - g) + \alpha(\alpha - 1)sAK_{LR}^{\alpha-1}\hat{k}. \\ &= \alpha(\alpha - 1)(g + \delta)\hat{k} \\ &= (\alpha - 1)(\delta + g)\hat{y}.\end{aligned}$$

- So we get

$$\nu = (g + \delta)(1 - \alpha).$$